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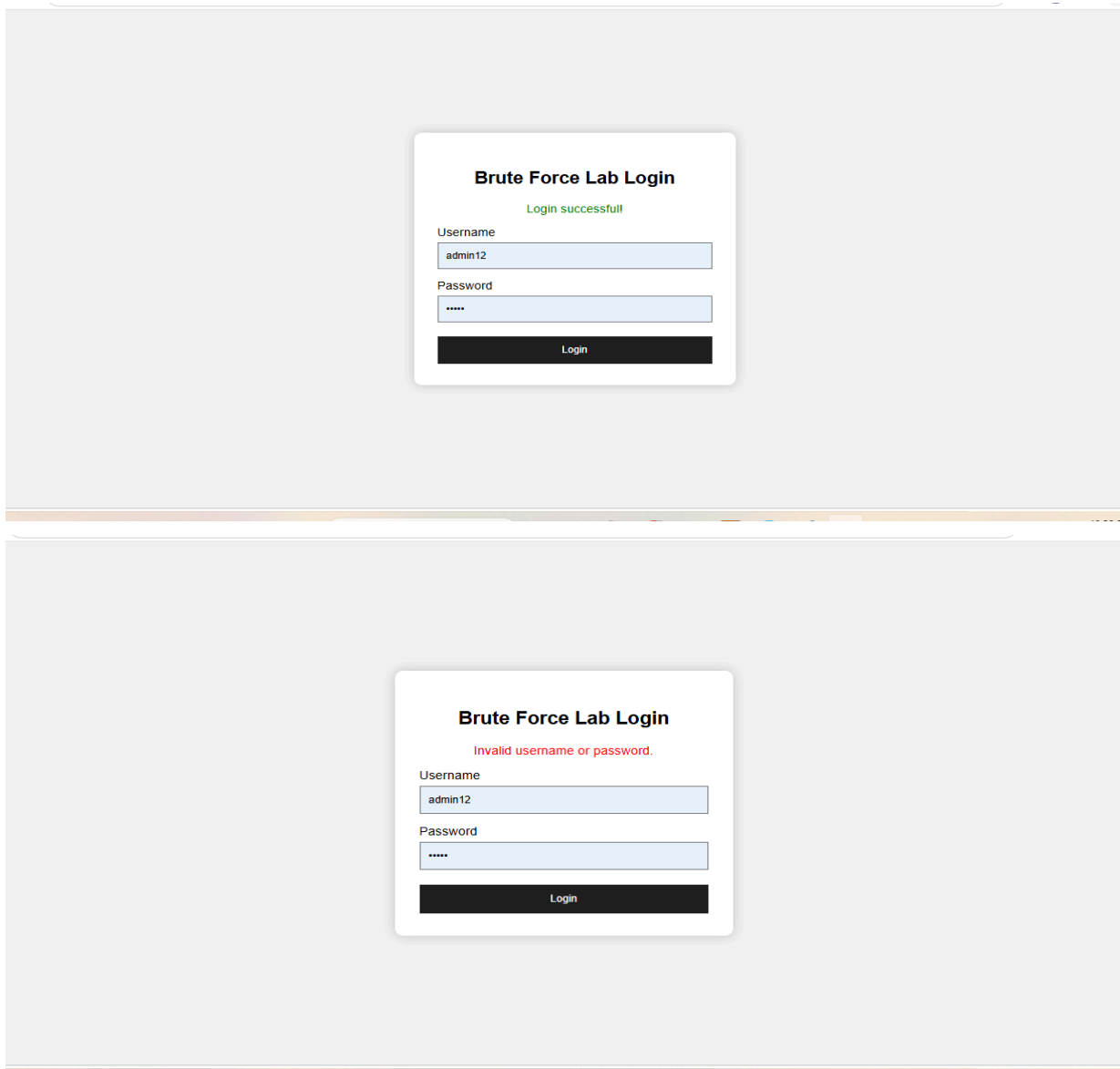
Date: Sept. 15, 2026

Course/Section: IT-107(IPSN3)

Activity 2: Authentication & Rate-Limiting Laboratory Analysis

This laboratory activity compares unprotected and protected login forms. It demonstrates how brute-force attacks can guess passwords and how temporary account lockouts can help prevent repeated failed attempts.

I. Execution of the brute-force test against the unprotected login form.



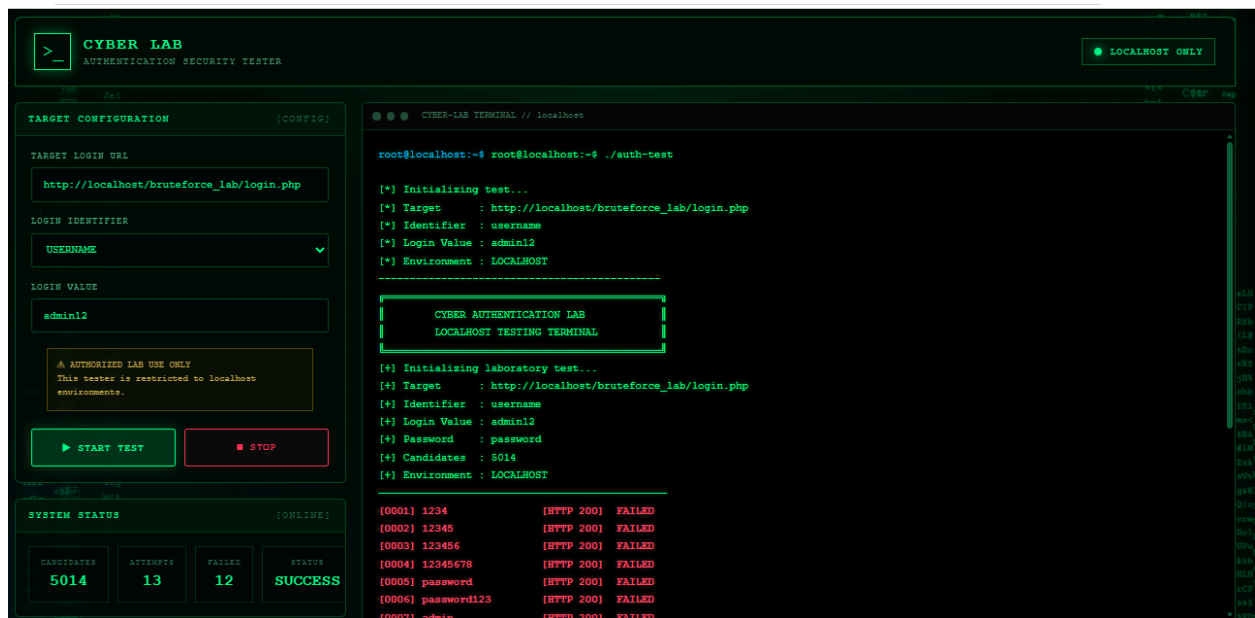
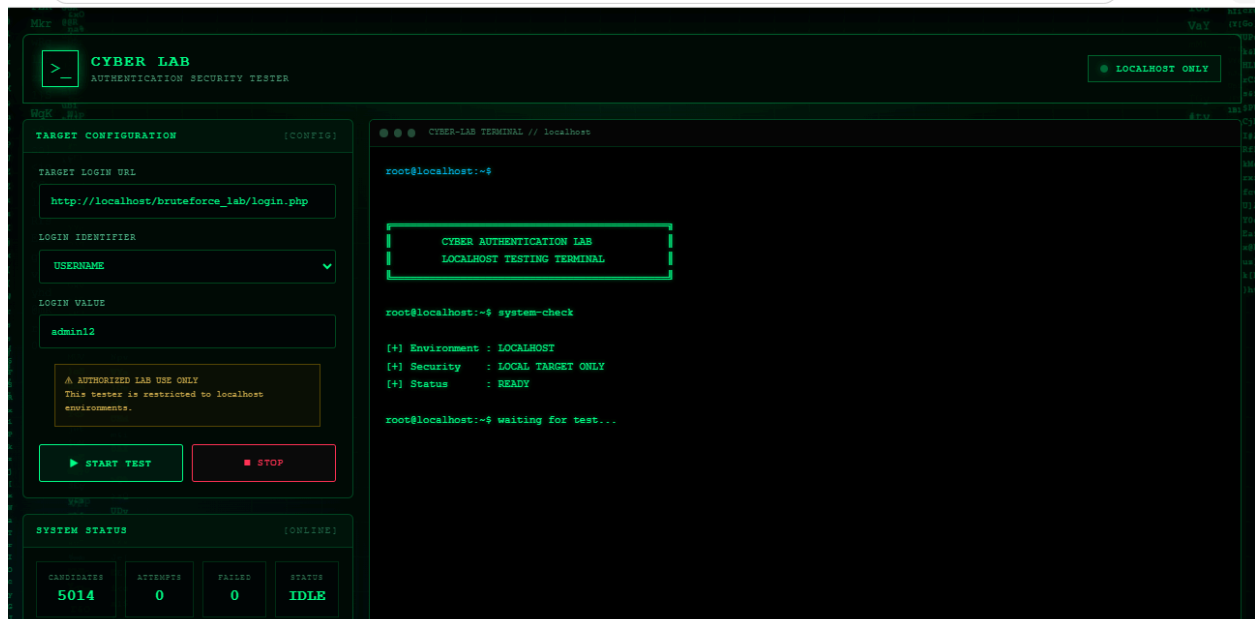
The image displays two screenshots of a web application's login interface, titled "Brute Force Lab Login".

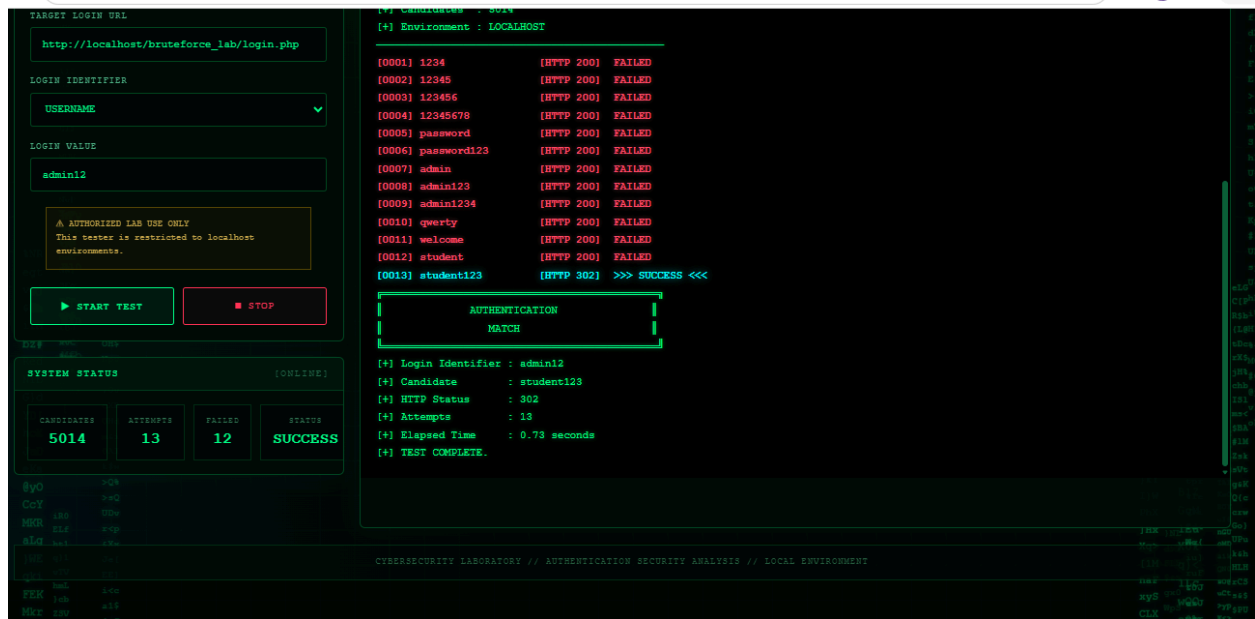
Top Screenshot (Successful Login):

- Message: Login successfull (in green text)
- Username field: admin12
- Password field: masked with asterisks (*****)
- Login button: visible at the bottom

Bottom Screenshot (Failed Login):

- Message: Invalid username or password. (in red text)
- Username field: admin12
- Password field: masked with asterisks (*****)
- Login button: visible at the bottom



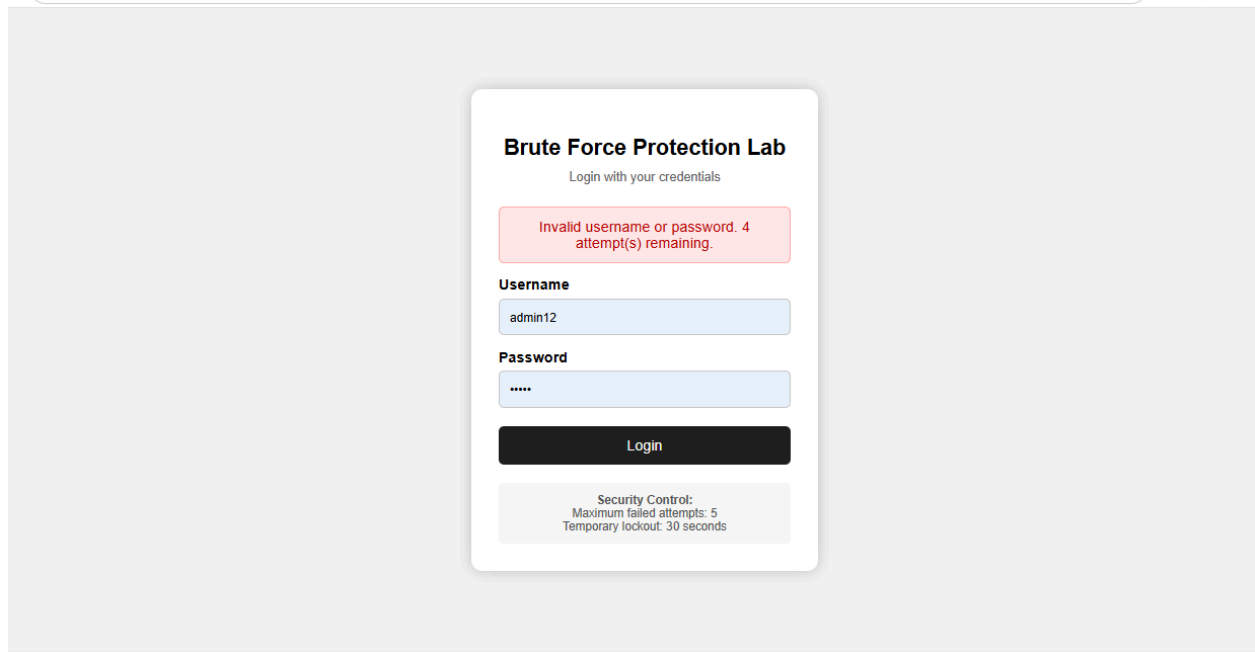


Execution Metrics - Unprotected Login

Metric	Result
Target	login.php
Username	admin12
Result	Password discovered
Discovered Password	student123
Total Attempts	13
Elapsed Time	0.73 seconds
HTTP Status	302

II. Protected Login Form

The login form was modified to include a temporary account lockout mechanism. The system allows a maximum of five consecutive failed login attempts. When the fifth failed attempt is reached, the login system temporarily blocks additional login requests for 30 seconds. Successful authentication resets the failed-attempt counter.



Brute Force Protection Lab
Login with your credentials

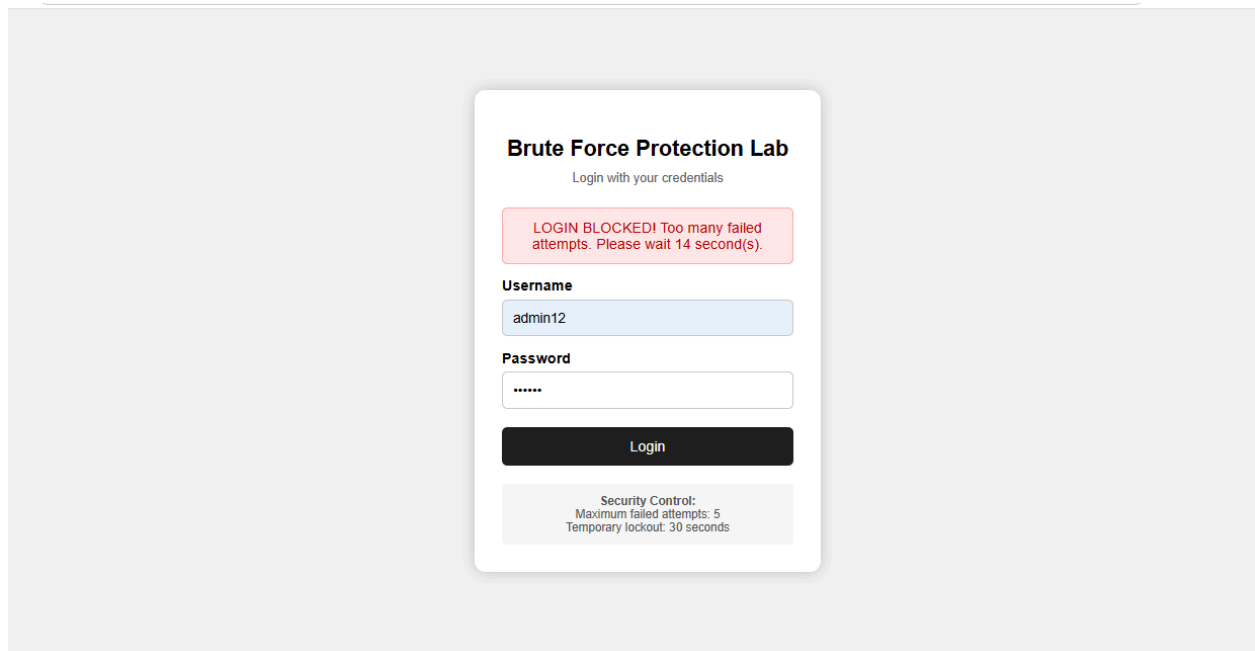
Invalid username or password. 4 attempt(s) remaining.

Username
admin12

Password

Login

Security Control:
Maximum failed attempts: 5
Temporary lockout: 30 seconds



Brute Force Protection Lab
Login with your credentials

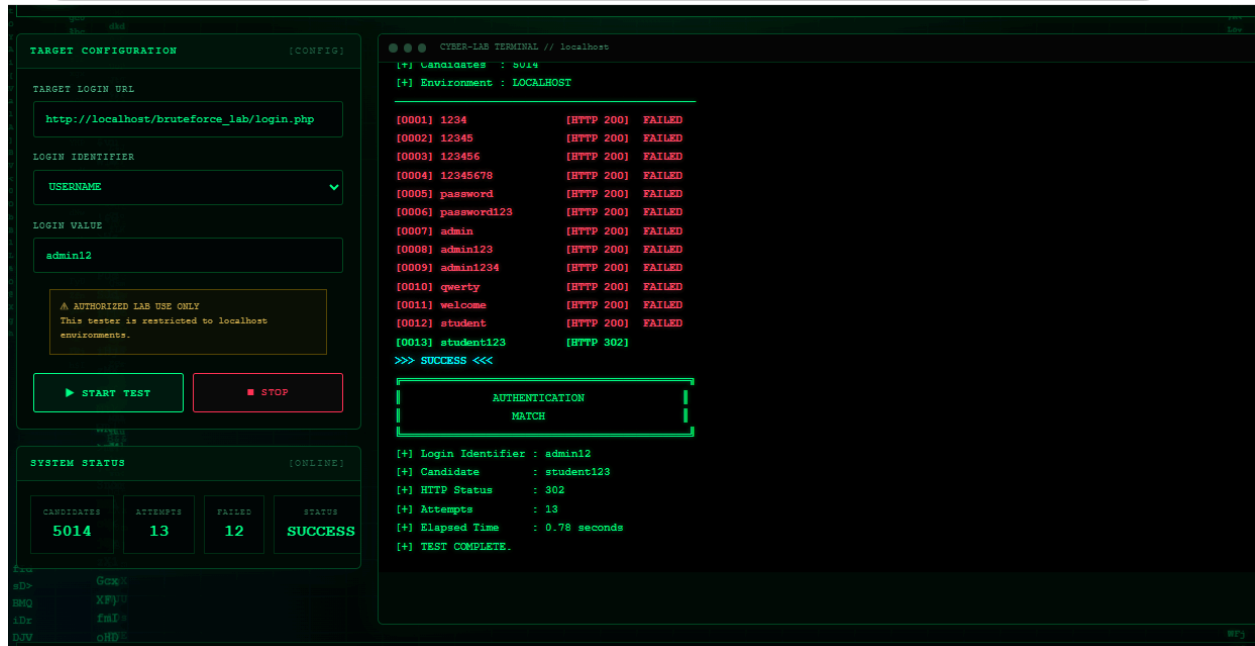
LOGIN BLOCKED! Too many failed attempts. Please wait 14 second(s).

Username
admin12

Password

Login

Security Control:
Maximum failed attempts: 5
Temporary lockout: 30 seconds



Execution Metrics - Protected Login

Metric	-Result
Target	login.php
Username	admin12
Maximum Failed Attempts	5
Lookout Duration	30 seconds
Result	Login blocked
Attempts Before Lockout	13
Elapsed Time	0.78 seconds
HTTP Status	302

III. Comparison of Both Tests

Metric	Unprotected	Protected
Brute-force testing	Allowed	Restricted
Failed-attempt limit	None	5 attempts
Lockout	None	30 seconds
Password discovered	Yes	Should be prevented during lockout
Automated requests	Continued	Blocked/Restricted
Security level	Weak	Improved

Analysis & Reflection

Rate Limiting: Rate limiting controls how often login attempts can be made. It helps slow down automated brute-force attacks and prevents attackers from making many password guesses quickly.

Account Lockouts: Account lockouts temporarily block login attempts after several failed attempts. In this activity, the account was locked after five failed attempts, preventing continuous password guessing.

CAPTCHA: CAPTCHA helps identify whether the login attempt is from a real person or an automated program. It makes automated brute-force attacks more difficult and works best when combined with rate limiting and account lockouts.